

SHARAD KUMAR

B.Tech Computer Science Student | Machine Learning & AI

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SUMMARY

B.Tech Computer Science student (3rd year, 9.4 CGPA) at Sant Longowal Institute of Engineering and Technology, with a strong Data Structures & Algorithms foundation from regular LeetCode/Codeforces practice. Currently focused on Machine Learning, with hands-on experience building and deploying a full ML pipeline end-to-end (data collection, feature engineering, model training/evaluation, deployment). Also learning FastAPI to serve ML models as backend services.

EDUCATION

Sant Longowal Institute of Engineering and Technology (SLIET)

Bachelor of Technology (B.Tech), Computer Science

Jun 2025 – May 2028

- CGPA: 9.4 (till current semester)

Diploma, Computer Science Engineering

Aug 2022 – May 2025

- CGPA: 9.2

PROJECTS

Codeforces Problem Difficulty Predictor | Python, scikit-learn, Pandas, Streamlit

- Built an end-to-end ML pipeline predicting a Codeforces problem's difficulty rating from its algorithm tags, using ~11,000 real problems pulled live from the Codeforces public API
- Engineered multi-label tag features and trained/compared three regression models (Ridge, Random Forest, Gradient Boosting); best model achieved a mean absolute error of ~454 rating points, with tags explaining ~35–40% of difficulty variance
- Deployed the trained model as a live, interactive web app using Streamlit, with the full pipeline (data fetch, training, app) version-controlled and documented on GitHub

[Live Demo](#) [GitHub](#)

Caissa – AI Chess Engine | Python, Pygame

- Collaborated with a team of 3 to build a full-featured chess engine and GUI in Python, supporting Player-vs-Player, Player-vs-AI, and AI-vs-AI modes
- Implemented complete chess rule logic (legal move generation, castling, en passant, pawn promotion, check/checkmate detection) with a modular, well-separated codebase (engine, AI, GUI, game loop)
- Built the AI opponent using Minimax search with Alpha-Beta pruning and a configurable search depth, plus a Pygame-based interface with board rendering and move animations

[GitHub](#)

SKILLS

- **Languages:** Python, JavaScript, HTML, CSS
- **ML / Data:** Machine Learning, scikit-learn, Pandas, Statistical Data Analysis
- **Backend / Tools:** FastAPI, Streamlit, Git, GitHub, Linux
- **Core CS:** Data Structures & Algorithms (active LeetCode / Codeforces practice)